Haris Naveed

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Education

New York University New York, New York
Master of Science in Data Science *May 2025*
- Fulbright Scholar

National University of Sciences and Technology (NUST) Islamabad, Pakistan
Bachelor's in Electrical Engineering *Aug. 2017 – June 2021*

Experience

Software Engineer - MLOps July 2025 – Present
Motive *Islamabad, Pakistan*
- Developed a langgraph - based agentic workflow to automate manual, repetitive pre and post deployment administrative tasks for a core service. Reduced time taken to build a new release from 81 to 25 minutes.
- Developed the backend logic for a service that leveraged Vision-Language Models (VLMs) to search thousands of videos to ones with identify unsafe driving behavior.

Graduate Teaching Assistant January 2025 – May 2025
NYU Center for Data Science *New York, New York*
- Graded assignments for DS - 1017: Responsible Data Science taught by Dr. Emily Black & Lucas Rosenblatt for the Spring '25 semester.
- Designed questions and updated homeworks to reflect the latest techniques used in auditing AI - based decision systems.
- Held weekly office hours to assist students with homework, projects and other questions.

Machine Learning Engineer September 2024 – Present
SimPPL *New York, New York*
- Developed a RAG (Retrieval-Augmented Generation) - based search system that accelerated narrative identification by producing an answer and retrieving its context from our closed-source data set.
- Set up the initial pipeline to process the raw data, generate embeddings and store them in a vector database, while ensuring that the system could handle large amounts of data in the future if necessary.
- Compared different RAG and retriever combinations on a set of factual, complex and vague questions. Identified GraphRAG with a multi-query retriever as the ideal system with an overall 60% win-rate.

Publications

M. H. Naveed, U. S. Hashmi, N. Tajved, N. Sultan and A. Imran, "Assessing Deep Generative Models on Time Series Network Data," in IEEE Access, vol. 10

Projects

Retrieval Augmented Generation for Financial Question-Answering | *Python, LangChain, ChromaDB*
- **Background:** Large language models (LLMs) often struggle with factual accuracy on queries. This is especially true for financial documents due to the presence of structured and unstructured data as well as specialized terminology.
- **Work:** Expanded on the **FinanceBench** paper by building a baseline model with an accuracy of 37%, and then exploring various techniques, including semantic chunking and contextual retrieval, to improve performance
- **Results:** Our experiments showed that a Naive RAG system combined with semantic chunking leads to an accuracy of 54%, a 17% improvement over the naive RAG baseline.

Skills

Programming Languages/Frameworks: Python, R, SQL, LangChain/Langgraph
Tools: Docker, K8s, Git, Linux, IntelliJ IDE, VS Code, RStudio, Claude Code
Libraries: Pandas, NumPy, matplotlib, PyTorch, scikit-learn, tidyverse, Fairlearn, statsmodels